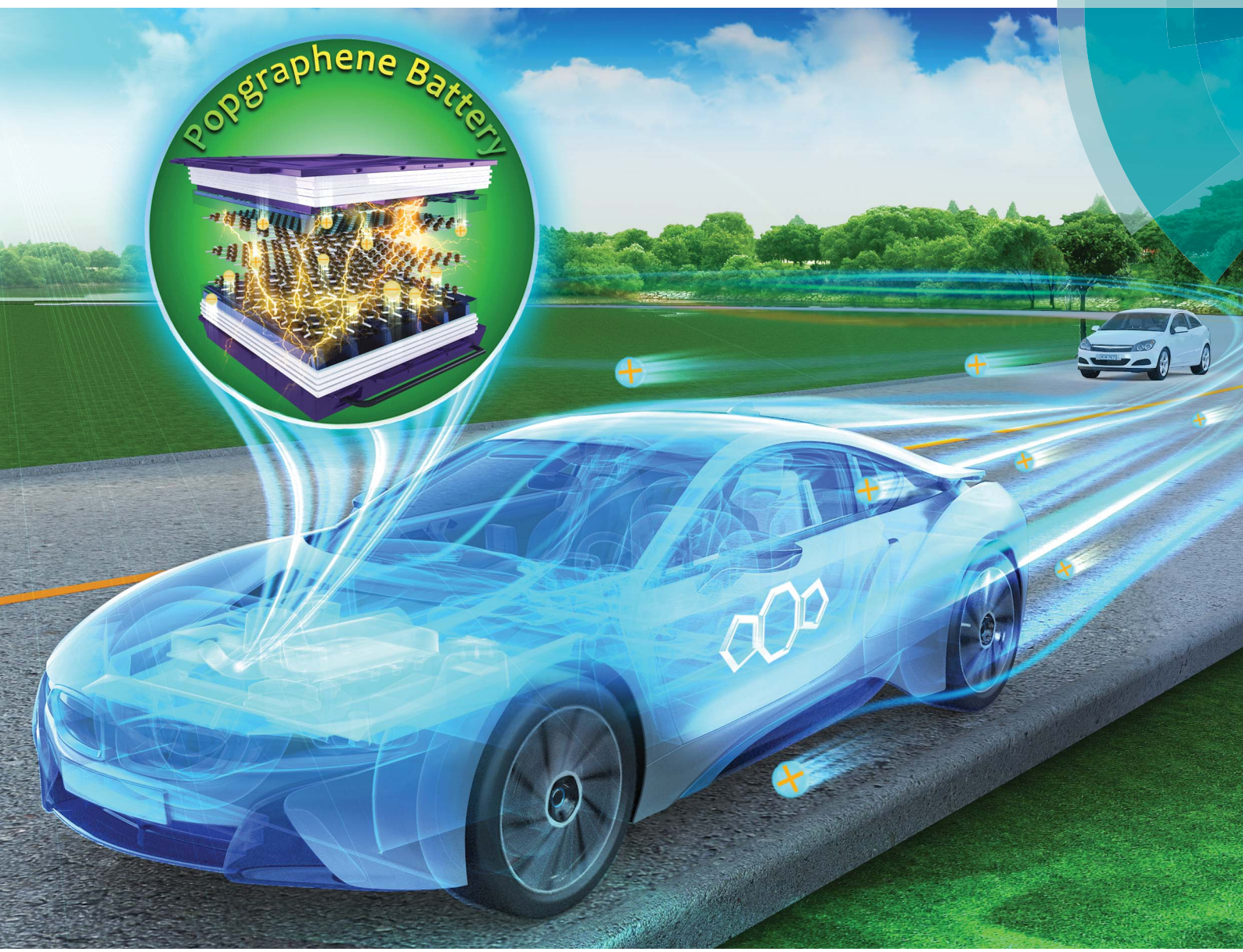
Volume 6 | Number 16 | 28 April 2018 | Pages 6691–7266

Journal of Materials Chemistry A

Materials for energy and sustainability

rsc.li/materials-a



ISSN 2050-7488

ROYAL SOCIETY OF CHEMISTRY

PAPER

Houyang Chen *et al.*

Popgraphene: a new 2D planar carbon allotrope composed of 5–8–5 carbon rings for high-performance lithium-ion battery anodes from bottom-up programming

Journal of Materials Chemistry A

PAPER

Published on 27 February 2018
Licence and permissions

Check for updates

Cite this: J. Mater. Chem. A, 2018, 6, 6815

Popgraphene: a new 2D planar carbon allotrope composed of 5–8–5 carbon rings for high-performance lithium-ion battery anodes from bottom-up programming†

Shuaiwei Wang, [a] Baocheng Yang,[a] Houyang Chen *[b] and Eli Ruckenstein [b]

Two-dimensional (2D) carbon allotropes have attracted great attention in both science and engineering fields. Their extraordinary and/or unique properties make them promising for engineering applications, ranging from metal-ion batteries to biosensors. Herein, by employing first-principles calculations, we propose a new 2D planar carbon allotrope, popgraphene, which is composed of 5–8–5 carbon rings. Using a bottom-up approach, popgraphene could be constructed *via* the formation of atomically precise 5–8–5 line defects in graphene by simultaneous electron irradiation. Popgraphene is intrinsically metallic. It has low energy and possesses great dynamic, thermal and mechanical stability. Our calculations also show that it has a high theoretical capacity for Li atoms (Li_4C_6: 1487 mA h g^{-1}), a low Li diffusion barrier (<0.55 eV), and a low average open circuit voltage (0.45 V). Such excellent characteristics suggest that programmable metallic popgraphene is a promising anode material for use in Li-ion batteries (LIBs) with fast charge/discharge rates. Our simulation results also indicate that van der Waals corrections play little role in the structural parameters of popgraphene, whereas they play a central role in the adsorption and diffusion abilities as well as the electrochemical performance of popgraphene. This study provides fundamental insight into the design of new materials with extraordinary properties for their excellent applications, including for the next generation of LIBs, and may open up a cutting-edge field in programming experimentally realized low-dimensional materials.

Received 14th January 2018
Accepted 26th February 2018

DOI: 10.1039/c8ta00438b

rsc.li/materials-a

1. Introduction

Carbon, which has attracted broad attention in both science and technology fields, exhibits numerous allotropes, such as 0D fullerene,[1] 1D carbon nanotubes,[2] 2D graphene,[2] 3D graphite[3] and 3D carbon honeycombs.[4–6] Among these structures, the classic 2D graphene material has remarkable electronic,[7–9] thermal,[10–12] and mechanical properties,[13] and exhibits the quantum Hall effect.[14] The giant success of graphene with its unique properties and extraordinary applications has encouraged scientists to explore new 2D carbon-based materials, such as penta-graphene,[15] phagraphene,[16] ψ-graphene,[17] and phographene.[18] However, unsuccessful experimental realization of these new proposed 2D carbon-based materials limits their further development. Thus, it is a grand challenge to develop new 2D carbon-based materials which can be realized experimentally.

The ultimate motivation for new materials development is their extraordinary and/or unique engineering application perspectives. 2D carbon-based materials can be used for numerous promising applications, ranging from biosensors,[19] to fuel cells,[20] to energy storage,[21] and to water purification.[22] One of the important applications is their use in metal-ion batteries (MIBs), especially for Li-ion batteries (LIBs). DFT results have revealed that Li can be stably adsorbed on penta-graphene (1489 mA h g^{-1}) with a low diffusion energy barrier,[23] however, a considerable band gap of 3.25 eV for penta-graphene[15] results in poor conductivity, which would limit its electrochemical performance. ψ-Graphene is a potential anode material for use in LIBs,[17] but its specific capacity (372 mA h g^{-1}) is much lower than that of penta-graphene. Phagraphene exhibits a good performance as an anode material, and its specific capacity (557.87 mA h g^{-1}) is rather high, however, obvious distortion occurs when more Li is adsorbed on the phagraphene.[24] In addition, both theoretical[25] and experimental[26] results have revealed the poor adsorption ability of Li atoms on pristine graphene, which limits its application for LIBs. Several studies have been performed to improve the

[a]Henan Provincial Key Laboratory of Nanocomposites and Applications, Institute of Nanostructured Functional Materials, Huanghe Science and Technology College, Zhengzhou 450006, China

[b]Department of Chemical and Biological Engineering, State University of New York at Buffalo, Buffalo, New York 14260-4200, USA. E-mail: hchen23@buffalo.edu

† Electronic supplementary information (ESI) available: Band structures, DOS, adsorption structures and diffusion barriers. See DOI: 10.1039/c8ta00438b

adsorption ability of Li atoms by creating active sites, such as divacancies[27,28] and Stone–Wales/5–7 rings defects.[27–29] Previous theoretical and experimental results indicate that non-hexagonal ring structures in 2D carbon allotropes would effectively improve and promote the adsorption ability of metal atoms such as Li, Na, Mg and Zn, which is a precondition for MIBs. The development of a 2D carbon allotrope that exhibits high storage capacity, good conductivity, a low diffusion energy barrier for metal ions and little structural change when metal atoms are adsorbed remains a huge challenge.

In a previous experiment, an atomically precise 5–8–5 line defect in graphene was prepared by simultaneous electron irradiation.[30] Based upon this procedure, we believe that it is possible to grow large area and highly ordered 5–8–5 defects in graphene. Thus, in this work, by employing a bottom-up approach, we have programmed atomically precise 5–8–5 defects in graphene and proposed a new 2D planar carbon allotrope, popgraphene, which was constructed from 5–8–5 carbon rings. Our DFT calculations show that metallic popgraphene has low energy and possesses dynamic, thermal and mechanical stability. Our results also reveal that it possesses a high Li capacity and fast Li mobility as well as a low open circuit voltage. These characteristics suggest its promising application as a high-performance anode material for use in LIBs.

2. Computational methods

All first-principles calculations were carried out using the Vienna *ab initio* simulation package (VASP).[31,32] Plane waves with an energy cutoff of 500 eV were employed and the energy precision was 10^{-6} eV. The ion–electron interaction was described by the projector-augmented plane wave (PAW) pseudopotential.[33] The generalized gradient approximation (GGA)[34] with Perdew–Burke–Ernzerhof (PBE) functional[35] was adopted to compute the electron exchange–correlation energy, while the Heyd–Scuseria–Ernzerhof (HSE06) functional[36] was used to calculate accurate electronic structures. van der Waals (vdW) interactions were taken into account by using a semiempirical correction of Grimme's D2 scheme.[34] A vacuum space of 15 Å in the direction normal to the sheet was employed to avoid interactions between the two layers. The structure was relaxed without any symmetry constraint. By using the conjugate gradients (CG) method, the lattice constants and atomic positions were fully relaxed until the maximum force on each atom was less than 0.01 eV $Å^{-1}$. The Brillouin zone was meshed by 7 × 7 × 1 grids for geometry optimization, calculation of band structures, density of states and the electron localization function (ELF), and by 13 × 13 × 1 grids for self-consistent calculations.

Phonon dispersion analysis was performed using the Phonopy code,[37]which interfaces with the density functional perturbation theory (DFPT),[38] to examine the dynamic stability. A 4 × 2 × 1 supercell was adopted to calculate the atomic force and the dynamical matrix. *Ab initio* molecular dynamics (AIMD) simulations, under a constant temperature and constant volume (NVT) ensemble, were carried out to evaluate the thermal stability, and the temperature was controlled using the Nosé–Hoover thermostat.[39] The climbing image nudged elastic band (CI-NEB) method[40,41] was employed to study the mobility of the Li ions.

3. Results and discussion

3.1 Structure

The structure of popgraphene (penta–octa–penta-graphene) is constructed from 5–8–5 carbon rings (Fig. 1). Similar to graphene, its structure is composed of a 2D one-atom-thick planar sheet. Its unit cell contains 12 carbon atoms and its plane group is *P2mg* (plane group no. 7). The optimized lattice constants are $a = 3.6833$ Å and $b = 9.1124$ Å with the carbon atoms occupying two nonequivalent atomic Wyckoff positions of 4*j* (0.5, 0.077) and 8*q* (0.690, 0.326), denoted C_1 and C_2, respectively. The bond length of C_2–C_2 is approximately 1.46 Å, which is longer than that of C_1–C_2 (1.44 Å) and C_1–C_1 (1.41 Å). The total energy of popgraphene is −8.97 eV per atom (Table 1), which is lower than that of penta-graphene (−8.32 eV per atom)[15] and a little higher than that of graphene (−9.23 eV per atom),[16] phagraphene (−9.03 eV per atom)[16] and ψ-graphene (−9.07 eV per atom).[17] Table 1 lists the lattice parameters, bond lengths, and total energies for popgraphene, phagraphene, penta-graphene, ψ-graphene, and graphene. It should be mentioned that the calculated lattice parameters and bond lengths with and without vdW corrections are almost the same (Table 1), suggesting that the effect of vdW corrections on the structural parameters of popgraphene is subtle and negligible.

3.2 Structural stability

To evaluate the structural stability, the dynamic, thermal, and mechanical stabilities were investigated using phonon dispersions, AIMD simulations and elastic constants, respectively. Both phonon dispersions and the phonon density of states show no imaginary modes (Fig. 2a), confirming that popgraphene is dynamically stable. The AIMD results, which were run with a 2 × 2 × 1 supercell, reveal that the structure remains in its original configuration at 300 K for 10 ps (Fig. 2b), whereas it shows a little distortion at 1500 K (Fig. 2c), indicating that popgraphene is thermally stable at both room temperature and high temperature (∼1500 K or even higher). The calculated elastic constants C_{11}, C_{12}, C_{22}, and C_{66} are 342.1, 62.8, 275.7 and 100.1 GPa, respectively. They satisfy the criteria[42] for mechanical stability, *i.e.* $C_{11}C_{22} - {C_{12}}^2 > 0$ and $C_{66} > 0$.

3.3 Electronic properties

The electronic band structure and the corresponding total and partial density of states (DOS) were calculated to evaluate the electronic performance of the popgraphene structure. On the PBE level, no bandgap was found between the valence and conduction bands (Fig. 3a), indicating that popgraphene is metallic. Meanwhile, there exists a flat band around the Fermi level along the *X*–*Z* line and more than one steep band on the *Γ*–*X* line. According to previous studies,[43,44] the coexistence of a flat band and a steep band indicates that the material may be

 This journal is © The Royal Society of Chemistry 2018

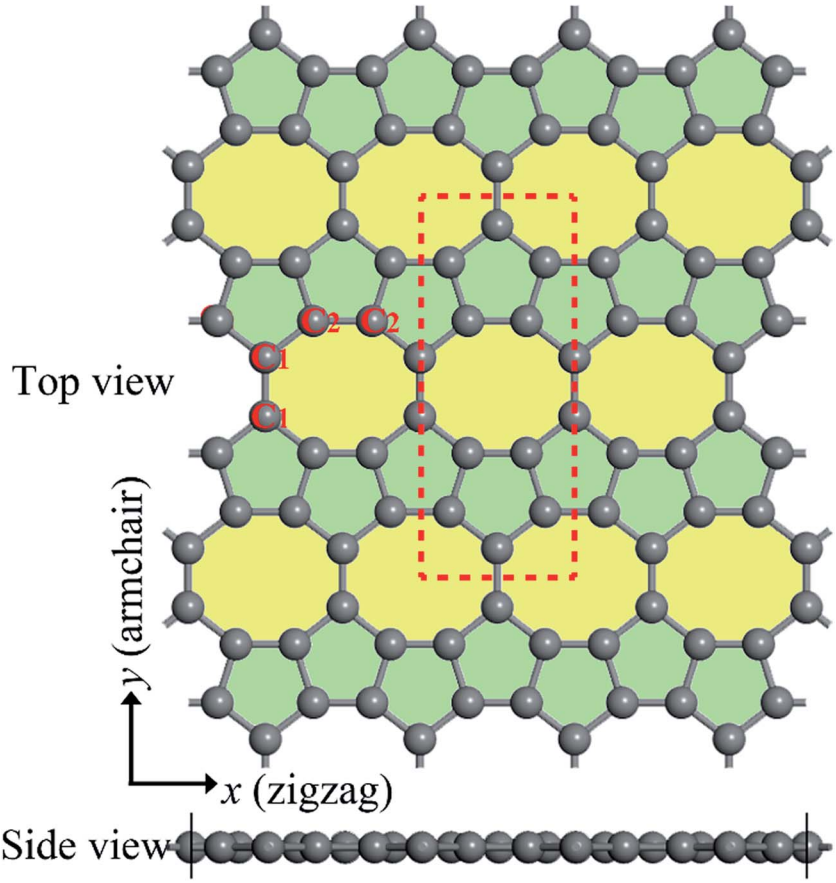


Fig. 1 The optimized structure of popgraphene, which includes van der Waals (vdW) corrections. The red dashed rectangle represents the unit cell of popgraphene. The *x* and *y* axes are along the zigzag and armchair directions, respectively.

a superconductor. Taking vdW corrections into account, the band structure was almost unchanged (Fig. S1†), indicating that the effect of vdW corrections on the band structure is negligible. Since the PBE functional usually underestimates the bandgap, the HSE06 hybrid functional, which can provide more accurate gap values, was employed to calculate the band structure. The band structures, calculated from the HSE06 hybrid functional (Fig. S2†), exhibit similar characteristics to those calculated from the PBE functional (Fig. 3a), and popgraphene remained metallic. The direction-dependent electronic property was further investigated under uniaxial and biaxial loading. By increasing the strain up to ±5% in the armchair, zigzag, and biaxial directions, popgraphene remained metallic (Fig. S3†). It should be noted that +5% strain occurs under tensile loading, whereas −5% strain happens under compression. To further determine the origin of the metallic state of popgraphene, its total and partial DOS are shown in Fig. 3b. Again, the non-zero value of the total DOS around the Fermi level confirms its metallic state. The large peaks for the total DOS around the Fermi level indicate the outstanding electronic transport ability of popgraphene. The total DOS around the Fermi level (−2 to 2 eV) is mainly attributed to the electrons of the p_z orbitals, and the contributions of the s, p_x and p_y orbitals are negligible.

Table 1 The calculated equilibrium structural parameters (lattice constants *a* and *b*, and bond lengths d_{C-C}), and total energy E_{tot} of popgraphene. For comparison, the corresponding parameters for phagraphene, penta-graphene, ψ-graphene and graphene are also provided

Structure		*a* (Å)	*b* (Å)	d_{C-C} (Å)	E_{tot} (eV per atom)
Popgraphene	Without vdW corrections	3.68	9.11	1.40–1.46	−8.91
	With vdW corrections	3.68	9.10	1.40–1.46	−8.97
Phagraphene[a]		8.09	6.65	1.40–1.52	−9.03
Penta-graphene[b]		3.64	3.64	1.34, 1.55	−8.32
ψ-Graphene[c]		6.70	4.84	1.41–1.51	−9.07
Graphene		2.46[d]	2.46[d]	1.42[d]	−9.23[e]

[a] Ref. 16. [b] Ref. 15. [c] Ref. 17. [d] Ref. 3. [e] Ref. 16.

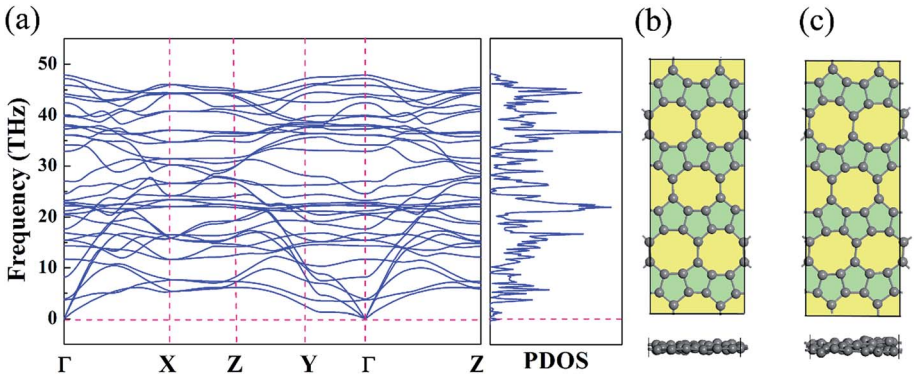


Fig. 2 (a) The phonon band structures and the corresponding DOS of a popgraphene sheet. The equilibrium structures of popgraphene during AIMD simulations at (b) 300 and (c) 1500 K.

We then analyzed the electron localization function (ELF) to understand the bonding nature. The values for the ELF are between 0.0 and 1.0. Values of 1.0, 0.5 and ~0 indicate perfect localization of electrons, fully delocalized electrons, and low charge density, respectively. As shown in Fig. 3c, the localization of electrons between the carbon atoms suggests strong σ-bonding states between neighboring carbon atoms. Combined with the calculated partial DOS results, one can conclude that the σ-bonding states are generated by the hybridization of s, p_x, and p_y orbitals, whereas the delocalized π bonds are formed by the p_z orbitals.

3.4 Promising applications of popgraphene as an anode material for LIBs

To assess the application of popgraphene for LIBs, the adsorption energy, storage capacity, possible volume change and mobility were taken into account.

The adsorption ability of Li ions is the fundamental condition for popgraphene to be considered as an anode material for LIBs. The adsorption energy E_{ads}, which was employed to assess the adsorption behavior of Li ions on popgraphene, is defined as $E_{ads} = (E_{n\text{Li–pop}} - E_{pop} - nE_{Li})/n$. $E_{n\text{Li–pop}}$ and E_{pop} are the total energies of the popgraphene structure after and before Li atom adsorption, E_{Li} represents the total energy per Li atom in bulk metal Li, and n is the number of Li atoms. First, we investigated the adsorption behavior of a single Li atom on the surface of popgraphene. All possible adsorption sites (two hollow sites: A, B; four bridge sites: C–F; and two top sites: G, H) were taken into account (Fig. 4a). After structural optimization, four adsorption sites (A–D) remained, while the others changed to the hollow site A or B. Taking vdW corrections into account, the adsorption energies of a single Li atom on A, B, C and D sites are −0.95, −0.71, −0.43, and −0.57 eV, respectively, indicating that they are exothermic and energetic processes. It should be noted that, without vdW corrections, the adsorption energies are −0.62, −0.53, −0.29, and −0.37 eV, which are higher than those with vdW corrections. This demonstrates that (1) the most favorable position is the hollow site A, and (2) the Li atoms are preferentially located on the hollow sites rather than the bridge sites. Furthermore, the adsorption energies of these four sites are close, showing the good mobility of Li atoms on the popgraphene sheet.

We further computed the storage capacity, which is a crucial factor for assessing the effective application of an anode

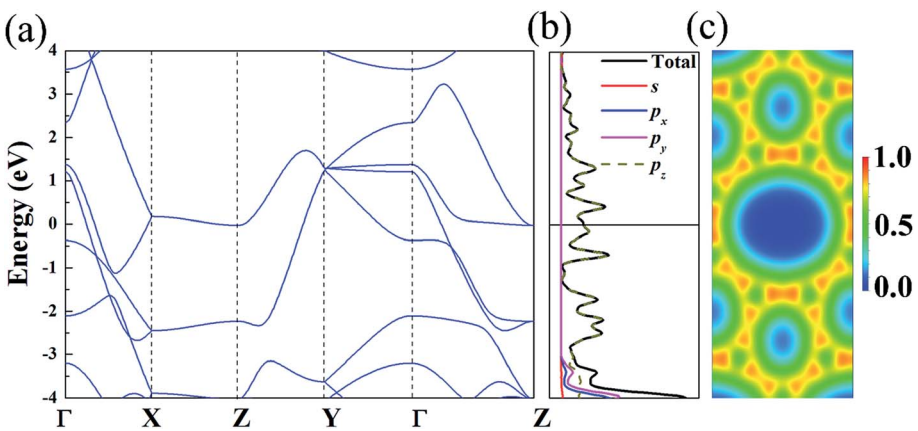


Fig. 3 (a) The band structures and (b) the projected DOS of a popgraphene sheet, and (c) the ELF map of popgraphene, with a slice crossing the structure plane, based on the PBE functional without considering vdW corrections.

This journal is © The Royal Society of Chemistry 2018

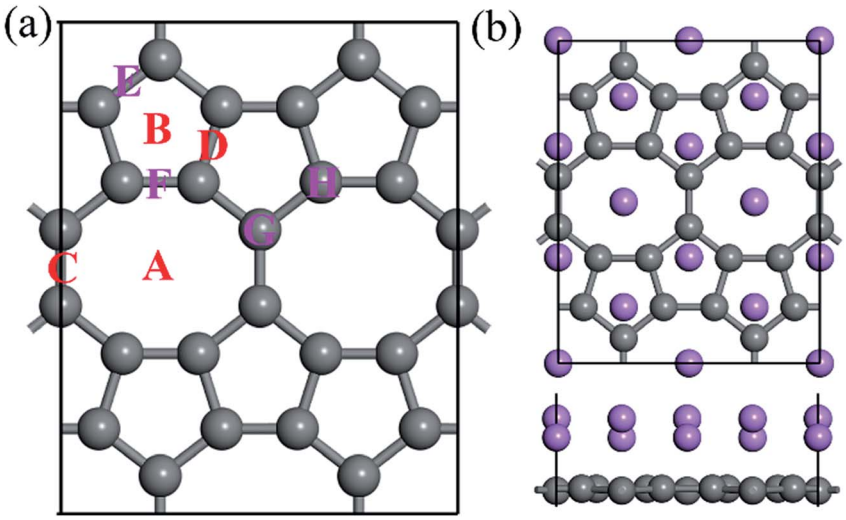


Fig. 4 (a) 2 × 1 supercell of a popgraphene sheet. The letters indicate the possible Li adsorption sites. (b) The top and side views of a popgraphene sheet with twelve Li adsorbed based on vdW corrections. The purple and gray balls represent the lithium and carbon atoms, respectively.

material for use in a battery. We added Li atoms one by one in the hollow and bridge sites of the 2 × 1 cell of a popgraphene sheet to determine the preferred adsorption sites on one side, until all the possible positions were considered. Our DFT results, which include vdW corrections, show that when twelve Li atoms accumulated on the popgraphene sheet, the adsorption energy was −0.15 eV per Li. The twelve Li atoms were preferentially located on the hollow sites A and B (Fig. 4b), and no visible distortion occurred in the popgraphene sheet. The corresponding theoretical storage capacity of the popgraphene sheet (Li_3C_6) was 1115.73 mA h g^{-1}. When both sides were taken into account, the maximal adsorption capacity was sixteen Li atoms with an adsorption energy of −0.12 eV (see Fig. S4†), and the corresponding theoretical capacity became Li_4C_6 and its value approached 1487.65 mA h g^{-1}. Our AIMD simulation results show that the structure of popgraphene with lithium ions, Li_4C_6, is thermally stable (Fig. S5†). The storage capacity of popgraphene is much larger than that of most of the other carbon anode materials (*e.g.*, 372 mA h g^{-1} (ref. 45 and 46) for graphite, 557.87 mA h g^{-1} (ref. 24) for phagraphene, and 372 mA h g^{-1} (ref. 17) for ψ-graphene), but is lower than that of graphene with maximum divacancies and Stone–Wales defects (1675 mA h g^{-1}).[27] In addition, the theoretical capacity without considering vdW corrections was also calculated. Our results reveal that the maximum theoretical capacity was 1115.73 mA h g^{-1} (Li_3C_6). Accordingly, vdW corrections play a significant role in the adsorption ability and storage capacity calculations and the results that include vdW corrections show a higher storage capacity than those calculated without considering vdW corrections.

The mobility of Li ions on the popgraphene sheet is another significant factor for the evaluation of its application as an anode material for use in LIBs. Four possible paths for the movement of the Li ion are considered (Fig. 5a), and their

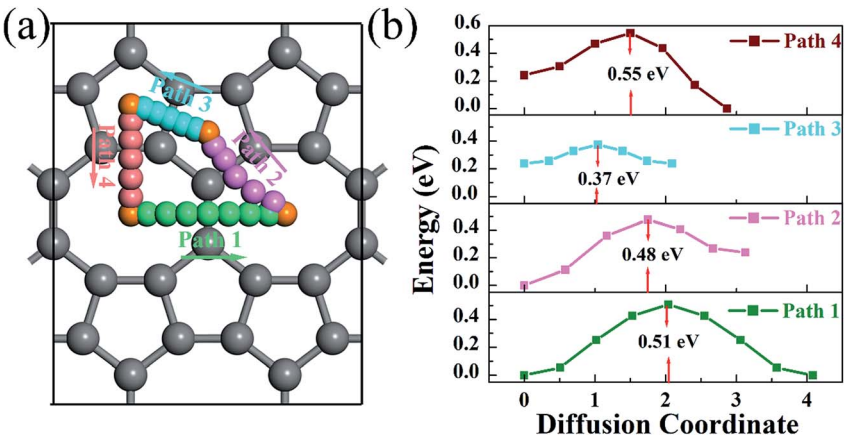


Fig. 5 Four possible migration paths (a) and their corresponding diffusion energy profiles (b) for Li diffusion on a popgraphene sheet, which includes vdW corrections.

diffusion barriers are 0.51/0.31, 0.48/0.30, 0.37/0.24 and 0.55/0.36 eV (with/without vdW corrections) (Fig. 5b/Fig. S6†), respectively. These barriers are comparable to those of phagraphene (0.08–0.42 eV)[24] and graphite (0.218–0.4 eV).[47] The lower barrier led to higher Li ion mobility, indicating that popgraphene possesses fast charge–discharge rates in the battery.

Finally, the average open circuit voltage (OCV) for Li intercalation on popgraphene was computed. The OCV was calculated from the energy difference based on the equation:[48] $\mathrm{OCV} \approx [E_{\mathrm{Li}_{x_1}-\mathrm{pop}} - E_{\mathrm{Li}_{x_2}-\mathrm{pop}} + (x_2 - x_1)E_{\mathrm{Li}}]/(x_2 - x_1)e$ when both volume and entropy effects were neglected. $E_{\mathrm{Li}_{x_1}-\mathrm{pop}}$ and $E_{\mathrm{Li}_{x_2}-\mathrm{pop}}$ are the total energies of Li_{x_1}–popgraphene and Li_{x_2}–popgraphene, respectively. The OCV values for the LiC_6, Li_2C_6, Li_3C_6, and Li_4C_6 systems are 0.50/0.18, 0.33/0.02, 0.15/0.01 and 0.12/NAN V (with/without vdW corrections), respectively. The calculated average OCV is 0.45/0.19 V (with/without vdW corrections) which is higher than that of graphite[49] (0.11 V). A low OCV results in a high operating voltage and a large capacity in LIBs. In short, the extremely high Li ion capacity, low OCV, fast charge–discharge rates and excellent electronic conductivity, suggest that popgraphene could be a highly effective anode material for use in LIBs.

4. Conclusions

By employing first-principles calculations, popgraphene, a new 2D planar carbon network composed of 5–8–5 carbon rings, which could be realized experimentally using simultaneous electron irradiation, was proposed and identified. The programmable metallic popgraphene, which was designed using a bottom-up method, has a low energy structure. It possesses dynamic, thermal and mechanical stability. Its high adsorption capacity (1487 mA h g^{-1} and Li_4C_6), low diffusion barrier (<0.55 eV), low average open circuit voltage (~0.45 V), and its metallic structure with excellent conductivity make it a promising and extraordinary anode material for use in LIBs. Our simulation results also indicate that the structural parameters of popgraphene are insensitive to vdW corrections, whereas the adsorption and diffusion abilities as well as the electrochemical performance of popgraphene are sensitive to vdW corrections. This work may provide a new strategy for programming experimentally realized low-dimensional materials and could open up a new field in designing next generation electrode materials for use in metal-ion batteries.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work was supported by the University at Buffalo, the National Natural Science Foundation of China (NSFC) (Grant No. 21206049) and the Innovation Scientists and Technicians Team Construction Projects of Henan Province (yukerenzu [2017]1).

References

1 H. W. Kroto, J. R. Heath, S. C. O'Brien, R. F. Curl and R. E. Smalley, *Nature*, 1985, **318**, 162–163.
2 K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, Y. Zhang, S. V. Dubonos, I. V. Grigorieva and A. A. Firsov, *Science*, 2004, **306**, 666–669.
3 F. Tuinstra and J. L. Koenig, *J. Chem. Phys.*, 1970, **53**, 1126–1130.
4 N. V. Krainyukova and E. N. Zubarev, *Phys. Rev. Lett.*, 2016, **116**, 055501.
5 S. Wang, D. Wu, B. Yang, E. Ruckenstein and H. Chen, *Nanoscale*, 2018, **10**, 2748–2754.
6 Z. Pang, X. Gu, Y. Wei, R. Yang and M. S. Dresselhaus, *Nano Lett.*, 2017, **17**, 179–185.
7 A. K. Geim, *Science*, 2009, **324**, 1530–1534.
8 A. K. Geim and K. S. Novoselov, *Nat. Mater.*, 2007, **6**, 183–191.
9 K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, M. I. Katsnelson, I. V. Grigorieva, S. V. Dubonos and A. A. Firsov, *Nature*, 2005, **438**, 197–200.
10 A. A. Balandin, S. Ghosh, W. Bao, I. Calizo, D. Teweldebrhan, F. Miao and C. N. Lau, *Nano Lett.*, 2008, **8**, 902–907.
11 Y. Liu, J. Huang, B. Yang, B. G. Sumpter and R. Qiao, *Carbon*, 2014, **75**, 169–177.
12 Y. Liu, C. Hu, J. Huang, B. G. Sumpter and R. Qiao, *J. Chem. Phys.*, 2015, **142**, 244703.
13 C. Lee, X. Wei, J. W. Kysar and J. Hone, *Science*, 2008, **321**, 385–388.
14 Y. Zhang, Y.-w. Tan, H. L. Stormer and P. Kim, *Nature*, 2005, **438**, 201.
15 S. Zhang, J. Zhou, Q. Wang, X. Chen, Y. Kawazoe and P. Jena, *Proc. Natl. Acad. Sci. U. S. A.*, 2015, **112**, 2372–2377.
16 Z. Wang, X.-F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao and A. R. Oganov, *Nano Lett.*, 2015, **15**, 6182–6186.
17 X. Li, Q. Wang and P. Jena, *J. Phys. Chem. Lett.*, 2017, **8**, 3234–3241.
18 X. Fan, J. Li and G. Chen, *RSC Adv.*, 2017, **7**, 17417–17426.
19 Y. Wang, Y. Shao, D. W. Matson, J. Li and Y. Lin, *ACS Nano*, 2010, **4**, 1790–1798.
20 X. Wang, L. Zhi and K. Müllen, *Nano Lett.*, 2008, **8**, 323–327.
21 R. Raccichini, A. Varzi, S. Passerini and B. Scrosati, *Nat. Mater.*, 2014, **14**, 271.
22 M. Bartolomei, E. Carmona-Novillo, M. I. Hernández, J. Campos-Martínez, F. Pirani, G. Giorgi and K. Yamashita, *J. Phys. Chem. Lett.*, 2014, **5**, 751–755.
23 B. Xiao, Y.-c. Li, X.-f. Yu and J.-b. Cheng, *ACS Appl. Mater. Interfaces*, 2016, **8**, 35342–35352.
24 D. Ferguson, D. J. Searles and M. Hankel, *ACS Appl. Mater. Interfaces*, 2017, **9**, 20577–20584.
25 E. Lee and K. A. Persson, *Nano Lett.*, 2012, **12**, 4624–4628.
26 E. Pollak, B. Geng, K.-J. Jeon, I. T. Lucas, T. J. Richardson, F. Wang and R. Kostecki, *Nano Lett.*, 2010, **10**, 3386–3388.
27 D. Datta, J. Li, N. Koratkar and V. B. Shenoy, *Carbon*, 2014, **80**, 305–310.
28 D. Datta, J. Li and V. B. Shenoy, *ACS Appl. Mater. Interfaces*, 2014, **6**, 1788–1795.

 This journal is © The Royal Society of Chemistry 2018

29 Z. Pang, X. Shi, Y. Wei and D. Fang, *Carbon*, 2016, **107**, 557–563.
30 J.-H. Chen, G. Autès, N. Alem, F. Gargiulo, A. Gautam, M. Linck, C. Kisielowski, O. Yazyev, S. Louie and A. Zettl, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2014, **89**, 121407.
31 G. Kresse and J. Furthmüller, *Comput. Mater. Sci.*, 1996, **6**, 15–50.
32 G. Kresse and J. Furthmüller, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1996, **54**, 11169.
33 G. Kresse and D. Joubert, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1999, **59**, 1758.
34 S. Grimme, *J. Comput. Chem.*, 2006, **27**, 1787–1799.
35 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865.
36 J. Heyd, G. E. Scuseria and M. Ernzerhof, *J. Chem. Phys.*, 2003, **118**, 8207–8215.
37 A. Togo and I. Tanaka, *Scr. Mater.*, 2015, **108**, 1–5.
38 K. Parlinski, Z. Li and Y. Kawazoe, *Phys. Rev. Lett.*, 1997, **78**, 4063.
39 S. Nosé, *J. Chem. Phys.*, 1984, **81**, 511–519.
40 G. Henkelman, B. P. Uberuaga and H. Jónsson, *J. Chem. Phys.*, 2000, **113**, 9901–9904.
41 G. Henkelman and H. Jónsson, *J. Chem. Phys.*, 2000, **113**, 9978–9985.
42 Y. Ding and Y. Wang, *J. Phys. Chem. C*, 2013, **117**, 18266–18278.
43 Y. Ding, S. Deng and Y. Zhao, *J. Mod. Transp.*, 2014, **22**, 183–186.
44 C. Zhong, Y. Xie, Y. Chen and S. Zhang, *Carbon*, 2016, **99**, 65–70.
45 J.-M. Tarascon and M. Armand, *Nature*, 2001, **414**, 359–367.
46 K. Toyoura, Y. Koyama, A. Kuwabara, F. Oba and I. Tanaka, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 2008, **78**, 214303.
47 J. Liu, S. Wang and Q. Sun, *Proc. Natl. Acad. Sci. U. S. A.*, 2017, **114**, 651–656.
48 M. Aydinol, A. Kohan, G. Ceder, K. Cho and J. Joannopoulos, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1997, **56**, 1354.
49 Y. Jing, Z. Zhou, C. R. Cabrera and Z. Chen, *J. Phys. Chem. C*, 2013, **117**, 25409–25413.

This journal is © The Royal Society of Chemistry 2018